

Aditi Kathalay

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Work Experience

L&T Finance | *AI Engineer*
Mumbai/Pune, India

Jun 2024 – Present

- Currently leading the development of an **Agentic AI Personal Loan Co-Pilot** using Gemini + GPT, LangChain, and tool calling to automate credit underwriting and assist credit managers in loan decision-making.
- Leading development of an **AI-powered Voicebot** for two-wheeler ABND customers using **GPT-4 LLM** with telephony integration, automating engagement, improved conversions and reduced call center load.
- Built a **Machine Learning–driven Gemini Lifestyle Index Rating model** to assess housing quality, cutting low-quality loan approvals and saving field officers' time; delivered full-stack AI solution (frontend, backend, APIs), showcased at RAISE 2024.
- Developed a **Treasury Forecasting model** using ML (Prophet), improving product-level financial accuracy for treasury planning.

L&T Finance | *Data Science Intern* | Mumbai, India

Jan 2024 – May 2024

Skills

AI/ML: Machine Learning, Agentic AI- Tool/Function Calling, LangChain, LangGraph, LLMs, MCP, RAG

Programming: Python, R, SQL,

Frameworks and Tools: Flask API, Fast API, Power BI, Git, Excel

Cloud: Google Cloud Platform, AWS

Certifications

Python (Google–Coursera), SQL (UC Davis–Coursera), Tableau (Scaler)

Projects

Personalized Skincare and Beauty Assistant (Multi-Agent, MCP, RAG)

Ongoing

- Building a multi-agent conversational bot for skincare, haircare, and makeup, providing personalised recommendations powered by RAG with curated product data.
- Integrated **Model Context Protocol (MCP)** to link directly with e-commerce sites (Amazon, Flipkart, Nykaa), enabling seamless product suggestions and checkout.

AI-Powered Travel Planner (Multi-Agent, Streamlit, LangGraph)

2025

- Developed an interactive multi-agent travel planning tool (GitHub repo) that generates itineraries, hotels, attractions, weather, and cost breakdowns end-to-end.
- Designed streamlined agent workflows in LangGraph to prevent recursion loops, making the solution scalable for real-world API integration.

Diabetic Retinopathy Detection and Stage Classification (Python, OpenCV, Perceptron, CNN)

Aug 2023

- Applied GRAHAM preprocessing and trained CNN + Perceptron models to classify Diabetic Retinopathy stages, improving interpretability by visualizing key retinal features to support diagnostic adoption.

Education

M.Sc, Data Science and Big Data Analytics, Maharashtra Institute of Technology, Pune

Gold Medalist, GPA: 9.6

Aug 2022 – Jun 2024,

B.Sc, Computer Science, Savitribai Phule Pune University, Pune

Gold Medalist, GPA: 9.7

Aug 2019 – Jun 2022

Achievements

- Served as the **Master of Ceremonies** for the prestigious **L&T Finance Star Awards 2025** and received a company award for outstanding hosting.
- Presented research paper “*Towards Early Detection: A Deep Learning & AI Approach to Diabetic Retinopathy Diagnosis*” at the **International Conference on Machine Learning & Intelligent Systems (ICMIS 2023)**.